SMART INDIA HACKATHON • JUDGING DEFENSE CHEAT SHEET

AirTrace AI • Top 20 Expected Judge Questions & Defensible Answers

Comprehensive Technical, Physical, Engineering & Policy Defense Guide for Final Evaluation

Target Domain: AI Atmospheric Remote Sensing

Coverage: 20 Exhaustive Questions across 6 Categories

Verified Metrics: R^2 = 0.8899 • MAE = 16.83 µg/m³ • 28 Clusters

📁 Category: AI Architecture

Q01 AI ARCHITECTURE — MODEL DESIGN

Why did you choose a hybrid 1D-CNN + BiLSTM + Attention architecture instead of standard tabular models like XGBoost or pure Vision Transformers?

🎯 ***What the Judge is Testing:*** *Testing architectural justification and domain-specific inductive bias.*

Air quality is non-linear and governed by multi-day atmospheric accumulation and chemical reactions. Tabular models like XGBoost treat time steps as independent static columns, destroying temporal autocorrelation. Vision Transformers require massive datasets (millions of images) to avoid overfitting on daily grids.

Our hybrid architecture provides three distinct inductive biases:
1. **1D-CNN**: Acts as a localized chemical compressor, learning intra-day cross-gas ratios (NO2/CO, HCHO/NO2) across sliding 3-day kernels.
2. **Bidirectional LSTM**: Models 14-day temporal context in both forward and backward time, capturing multi-day pollutant accumulation and stagnant boundary-layer inversion.
3. **Self-Attention Pooling**: Dynamically assigns high attention weights (α_t) to severe episodic spike days rather than uniform temporal averaging.

Metric: Yields **R^2 = 0.8899** and **MAE = 16.83 µg/m³** (36.2% error reduction over standard LSTM baselines).

Q02 AI ARCHITECTURE — EXTREME TAILS

How does your model solve the 'regression-to-the-mean' collapse during severe winter smog episodes (>200 or >400 µg/m³)?

🎯 ***What the Judge is Testing:*** *Evaluating handling of skewed data distributions and extreme loss function design.*

Standard Mean Squared Error (MSE) heavily penalizes models toward the dataset mean (60–100 µg/m³), causing dangerous underestimations during winter peak pollution events in North India.

We developed a **Dual Regime Neural Blending Architecture**:
• **Raw Model (`cnn_lstm_raw`)**: Trained on standard distributions for baseline precision.
• **Extreme Tail Model (`cnn_lstm_extreme`)**: Trained with weighted tail loss focusing on the upper 15th percentile (>150 µg/m³).
• **Ensemble Optimization**: A systematic validation grid search identified the optimal blend: `ŷ = 0.925 × ŷ_raw + 0.075 × ŷ_extreme` .
Validation: Achieves a near-zero bias of **-0.19 µg/m³ on hazardous episodes >400 µg/m³** while preserving overall R^2 = 0.8899.

Q03 AI ARCHITECTURE — HYPERPARAMETERS

Why is the input sequence length fixed at 14 days (T=14) instead of 7 or 30 days?

🎯 ***What the Judge is Testing:*** *Testing hyperparameter experimentation and atmospheric residence time reasoning.*

We systematically benchmarked sequences of 3, 7, 14, 21, and 30 days:
• **7 Days**: Insufficient to capture the multi-day synoptic stagnation cycles and anticyclonic trapping over the Indo-Gangetic Plain.
• **30 Days**: Introduced excessive gradient variance and historical noise, particularly during monsoon airmass transition phases.
• **14 Days (Optimal)**: Perfectly matched the physical atmospheric residence time of fine particulate matter and precursor trace gases while keeping the input tensor compact at `(14 × 84)` .

Q04 AI ARCHITECTURE

FEATURE TENSOR

How do you transform 53 raw satellite features into an 84-dimensional preprocessed input tensor?

🎯 ***What the Judge is Testing:*** *Testing feature preprocessing pipeline and handling of missing values.*

The 53 raw features consist of 12 features each for CH4, CO, and NO2 (spatial mean, median, std, min, max, p10, p25, p75, p90, valid counts/fraction, point values), 4 features for HCHO, and 13 spatiotemporal/cyclical calendar features (lat, lon, sin/cos day of year, sin/cos month).

During preprocessing, a scikit-learn `MissingIndicator` appends boolean indicator masks for all satellite channels (tracking sensor coverage), and features are normalized via a `RobustScaler` fit on multi-year distributions. This expands the feature width from 53 raw variables to exactly **84 preprocessed tensor dimensions** over 14 timesteps `(Batch, 14, 84)`.

📁 Category: Remote Sensing & Physics

Q05 REMOTE SENSING & PHYSICS

COLUMN-TO-SURFACE

Satellites measure total vertical column density (molecules/cm^2). How can you legitimately convert this to surface-level respirable PM2.5 (µg/m^3)?

🎯 ***What the Judge is Testing:*** *Testing atmospheric physics and boundary layer understanding.*

Total column density integrates molecules from the ground to the top of the atmosphere. Direct linear conversion fails because ground concentrations depend on Planetary Boundary Layer Height (PBLH), relative humidity, and chemical conversion rates.

Our network learns non-linear physical conversion by coupling four multi-spectral channels:
1. **CO & NO2**: Serve as primary combustion and traffic emission proxies.
2. **HCHO**: Tracks secondary organic aerosol (SOA) gas-to-particle photochemical oxidation.
3. **CH4**: Accounts for synoptic airmass shifts.
4. **Cyclical Calendar Encodings**: `sin/cos(DOY)` implicitly model the seasonal cycle of boundary layer height collapse.
The BiLSTM-Attention layers learn how these column metrics accumulate into surface PM2.5, achieving **R^2 = 0.8899**.

Q06 REMOTE SENSING & PHYSICS — DATA RESILIENCE

How does the pipeline handle missing satellite data caused by heavy cloud cover or Monsoon rain?

🎯 ***What the Judge is Testing:*** *Testing data quality guardrails, fault tolerance, and hallucination prevention.*

We engineered a 3-layer automated data quality pipeline:
1. **Quality Masking**: We extract `valid_count` and `valid_fraction` for each raster pass. Fill values ≥30,000 and negative noise artifacts are strictly masked.
2. **Missingness Encoding**: The preprocessor incorporates a `MissingIndicator` mask alongside a fitted `SimpleImputer` trained on multi-year medians.
3. **Fail-Safe Validation Gate**: If fewer than 7 days in the 14-day window contain usable satellite data (`min_usable_days < 7`), the API refuses to generate hallucinated numbers and returns an explicit HTTP 400 `insufficient_satellite_history` error.

Q07 REMOTE SENSING & PHYSICS — SPATIAL RESOLUTION

Why is 'training_compat' (center ±0.05° bounding box) the default spatial mode in your prediction API?

🎯 ***What the Judge is Testing:*** *Testing understanding of spatial geometry extraction and feature distribution alignment.*

The production model was trained using a fixed ±0.05° bounding box (~11 km × 11 km) around monitoring stations to capture representative localized airsheds. If a user passes an arbitrary large polygon (e.g. 50 km × 50 km), the statistical distribution of percentiles (p10, p90) shifts significantly relative to the training distribution.

The API computes the geographic centroid of any input bounding box and reproduces the exact training geometry by default, guaranteeing feature distribution fidelity. An optional `SIH_SPATIAL_MODE=input_bbox` override is provided for custom regional aggregations.

Q08 REMOTE SENSING & PHYSICS — AOD INTEGRATION

Why is MOSDAC INSAT-3D AOD provided as an optional diagnostic rather than hardcoded into the 53-feature input tensor?

🎯 ***What the Judge is Testing:*** *Testing scientific honesty, model provenance, and version control.*

The trained model artifacts (weights, scaler, imputer) were trained on historical daily Sentinel-5P features where AOD was not part of the 53-feature matrix. Injecting uncalibrated AOD into fixed model weights would corrupt neural activations.

We built a production-grade **ISRO MOSDAC adapter service** (`mosdac_aod.py`) that dynamically ingests geostationary INSAT-3D AOD as an independent diagnostic layer. In Phase 2 of our roadmap, we are retraining a dedicated 57-feature tensor with unified spaceborne AOD assimilation.

📁 Category: HCHO & Wind Transport

Q09 HCHO & WIND TRANSPORT — SOURCE ATTRIBUTION

How does your platform mathematically differentiate between a true emission source and a downwind receptor area?

🎯 ***What the Judge is Testing:*** *Testing understanding of the physical transport model and wind vector calculations.*

Satellite columns aggregate all vertical molecules, meaning downwind receptor areas often appear 'polluted' despite producing zero emissions. We differentiate sources from receptors by coupling **TROPOMI HCHO columns with ECMWF ERA5 10m wind vectors (u_10, v_10) and MODIS Fire Radiative Power (FRP)**:
1. Derive wind speed and meteorological wind direction: `ws = sqrt(u^2 + v^2)` , `wd = (atan2(-u, -v) * 180/π + 360) mod 360` .
2. Compute candidate source anomaly scores: `S_i = [(C_i - μ_bg) / σ_bg] × [1 + ln(1 + FRP_i)]` .
3. Measure the **Downwind Enhancement Ratio**: `E_down = (C_downwind - C_upwind) / C_upwind` along the advection vector.
4. Conduct hypothesis testing: If enhancement is statistically significant (**p < 0.05**) and persistent (≥0.70), it is verified as a True Emission Source.

Q10 HCHO & WIND TRANSPORT — CLUSTERING

Why did you use HDBSCAN instead of K-Means or DBSCAN for spatial hotspot clustering?

🎯 ***What the Judge is Testing:*** *Testing clustering algorithm selection and spatial density understanding.*

• **K-Means** assumes spherical, uniformly sized clusters and forces all background noise cells into clusters, creating false pollution boundaries.
• **DBSCAN** requires a globally fixed density threshold (ε), which fails across India because agricultural burning in Punjab has vastly different spatial density than industrial clusters in Gujarat.
• **HDBSCAN** constructs a hierarchical density tree, allowing clusters of variable densities to be discovered while automatically identifying and filtering diffuse tropospheric background noise as unassigned noise points.
Result: Delineated **28 distinct, contiguous domestic Indian source regions** with 100% geographic traceability.

Q11 HCHO & WIND TRANSPORT — VALIDATION TIERS

What are your validation confidence tiers (Class A, Class B, Class C) and what do they signify?

🎯 ***What the Judge is Testing:*** *Testing regulatory classification criteria and categorization logic.*

Clusters are categorized into three defensible tiers based on source score, temporal persistence, and transport significance:
• **Class A (Strong Confidence)**: High composite source score, cluster persistence ≥ 0.70, statistically proven downwind enhancement ($p < 0.05$). *Examples: Sangli-Satara Sugar Belt (#34), Bharuch-Surat Petrochem (#12), Bhopal-Hoshangabad Valley (#26).*
• **Class B (Moderate Confidence)**: Consistent seasonal plume (0.40 ≤ persistence < 0.70). *Examples: Ludhiana-Jalandhar Belt (#1), Bilaspur-Korba Power Basin (#16).*
• **Class C (Weak / Diffuse)**: Lower persistence or diffuse regional agricultural background. *Examples: Bathinda-Sirsa (#0), Gorakhpur-Deoria Terai (#24).*

Q12 HCHO & WIND TRANSPORT — PHOTOCHEMISTRY

Why is Formaldehyde (HCHO) specifically chosen to study secondary aerosol (PM2.5) formation?

🎯 ***What the Judge is Testing:*** *Testing understanding of photochemical atmospheric mechanisms.*

Formaldehyde (HCHO) is a short-lived intermediate product resulting from the oxidation of Non-Methane Volatile Organic Compounds (NMVOCs). Because its atmospheric lifetime is brief (~hours), high HCHO concentrations indicate active, localized VOC emissions.

In the troposphere, HCHO photolysis produces hydroperoxyl radicals (HO2), which react with NO to form NO2 and hydroxyl radicals (OH), driving both **tropospheric ozone (O3) synthesis and Secondary Organic Aerosol (SOA / PM2.5) formation**. Tracking HCHO provides the earliest spaceborne indicator of secondary aerosol buildup.

📁 Category: Evaluation & Benchmarks

Q13 EVALUATION & BENCHMARKS — METRICS

Walk me through your quantitative evaluation results. How did you validate that your model is not overfitting?

🎯 ***What the Judge is Testing:*** *Testing validation rigor, data splitting, and performance benchmarks.*

The model was evaluated on an independent test split of **1,942 unseen station-day samples** across diverse geographical zones:
• **R^2 Score**: **0.8899** (explains 89% of ground variance vs 0.7140 on baseline LSTM).
• **Mean Absolute Error (MAE)**: **16.83 µg/m³** (vs 26.40 µg/m³ baseline).
• **Root Mean Squared Error (RMSE)**: **26.43 µg/m³** (vs 39.80 µg/m³ baseline).
• **Median Absolute Error**: **11.23 µg/m³**.
Overfitting was mitigated using Spatial Dropout, Batch Normalization, and early stopping on validation loss across temporal block splits.

Q14 EVALUATION & BENCHMARKS — STRESS TESTING

How does the model behave under future climate regime shifts or extreme stress testing?

🎯 ***What the Judge is Testing:*** *Testing domain generalization and model robustness.*

We conducted an out-of-distribution stress test across extreme unseen meteorological regimes (n=2,065). While standard models experience catastrophic failure during severe regime changes, our Dual Blending architecture maintains bounded error rates.

Furthermore, on the hazardous episode test set (>400 µg/m³), the blended model recorded an almost unbiased mean prediction of **430.47 µg/m³ against an actual mean of 430.66 µg/m³ (Bias = -0.19 µg/m³)**, demonstrating exceptional fidelity in emergency air quality regimes.

📁 Category: Software Architecture

Q15 SOFTWARE ARCHITECTURE — MICROSERVICES

Why did you build both a FastAPI backend and a Streamlit dashboard instead of a single monolithic script?

🎯 ***What the Judge is Testing:*** *Evaluating production readiness and software engineering maturity.*

A production environmental platform requires two distinct interfaces:
1. **FastAPI Backend (Port 8000)**: High-throughput, headless asynchronous REST API (`/predict` , `/health` , `/model-contract`) enabling automated machine-to-machine integrations, IoT sensor networks, and mobile app ingestion.
2. **Streamlit Frontend (Port 8501)**: Interactive visual decision-support dashboard designed for regulatory officers, policy analysts, and researchers with interactive Mapbox geospatial mapping, time-series curves, and CSV report downloads.

Q16 SOFTWARE ARCHITECTURE

SPEED & LATENCY

How did you optimize the dashboard to deliver sub-millisecond response times?

🎯 ***What the Judge is Testing:*** *Testing UI performance optimization and caching strategies.*

We applied four core engineering optimizations:
1. **Granular In-Memory Caching (`@st.cache_data`)**: Separated lightweight summary metadata (12 KB) from heavy point clouds, allowing instant initial page renders (<2 ms).
2. **On-Demand Lazy Loading**: Multi-megabyte HDBSCAN spatial grid points load exclusively when the user opens the grid tab.
3. **WebGL Hardware Acceleration**: Scatter plots use `render_mode='webgl'` to offload point-cloud rendering to the client GPU rather than rendering thousands of heavy SVG DOM nodes.
4. **Prediction Memoization**: Coordinate/date predictions are cached in RAM, making preset switches instantaneous (<1 ms).

📁 Category: Policy & Governance

Q17 POLICY & GOVERNANCE

ECONOMIC ROI

What is the economic business case and cost savings of AirTrace AI for environmental authorities?

🎯 ***What the Judge is Testing:*** *Evaluating commercial viability and public sector impact.*

Setting up and maintaining a single physical CAAQMS station costs ₹**1.5 Crore upfront and ₹25 Lakhs annually**. Monitoring all 700+ districts of India with physical ground hardware would cost over ₹**1,000+ Crores**.

AirTrace AI delivers 100% geographic coverage across India at near-zero hardware cost by unlocking open spaceborne Copernicus telemetry. It transforms satellite rasters into actionable, hyper-local intelligence, democratizing air quality monitoring for rural and underfunded municipal regions.

Q18 POLICY & GOVERNANCE

NCAP ALIGNMENT

How does your solution align with the National Clean Air Programme (NCAP) and CPCB mandates?

🎯 ***What the Judge is Testing:*** *Testing regulatory alignment and practical deployment feasibility.*

The National Clean Air Programme (NCAP) mandates a 40% reduction in particulate matter across 131 non-attainment cities by 2026. AirTrace AI directly supports NCAP through:
1. **Source Attribution**: Pinpoints whether pollution spikes are generated by localized industrial boilers (e.g. Sangli sugar belt) or transported stubble smoke.
2. **Actionable SPCB Directives**: Integrated NCAP Action Matrix links hotspot validation tiers directly to regulatory enforcement (e.g., CEMS audits, wet scrubbers, CRM machinery).
3. **Inter-State Dispute Resolution**: Provides objective, satellite-verified transport trajectories to resolve trans-boundary airshed disputes.

Q19 POLICY & GOVERNANCE

LEGAL EVIDENCE

Can your satellite-derived metrics be challenged in environmental courts by non-compliant industrial units?

🎯 ***What the Judge is Testing:*** *Testing legal defensibility and statistical robustness.*

Our transport verification is designed to meet strict evidentiary standards:
• We do not rely on raw optical columns alone; we calculate downwind enhancement ratios backed by **two-sample hypothesis testing ($p < 0.05$)**.
• We cross-correlate with independent ECMWF ERA5 wind vectors and MODIS/VIIRS thermal fire detections.
• Multi-month persistence scoring (≥0.70) proves chronic, systematic emissions rather than transient anomalies, providing regulators with statistically defensible audit dossiers.

📁 Category: Roadmap & Scalability

Q20 ROADMAP & SCALABILITY

FUTURE ROADMAP

What are your immediate next steps to scale AirTrace AI into a nationwide production platform?

🎯 ***What the Judge is Testing:*** *Evaluating long-term technical vision and operational readiness.*

Our 3-phase deployment roadmap comprises:

• **Phase 1 (Current - Completed)**: Fused 28 Indian HCHO hotspot clusters, calibrated Dual CNN-BiLSTM PM2.5 engine, and launched production FastAPI + WebGL dashboard.

• **Phase 2 (Q4 2026)**: Ingesting daily near-real-time (NRT) automated orbital streams with automated threshold-based WhatsApp/Email alert triggers for SPCB officers.

• **Phase 3 (2027)**: Ground-truth assimilation fusing live CPCB CAAQMS API feeds into the neural attention layer for continuous self-calibrating spatial prediction.